ARTIFICIAL INTELLIGENCE POWERED RESUME SCREENING BOT USING NLP & POWER BI

By Gurbani Kaur

TABLE OF CONTENTS

INTRODUCTION

This project is an AI-powered Resume Screening Bot that uses Natural Language Processing (NLP) to match resumes with a job description. It reads resumes, compares them to the job role, and ranks them based on textual relevance using TF-IDF and Cosine Similarity.

The backend is built in Python (Jupyter Notebook), and the results are shown in a Power BI dashboard — making the hiring process faster, smarter, and more efficient.

KEY TECHNOLOGIES USED:

- Python (Jupyter Notebook)
- NLP (TF-IDF, Cosine Similarity)
- Power BI (for result visualization)

PROJECT OBJECTIVES

OBJECTIVE 01

Build a tool that scores and ranks resumes based on keyword matching.

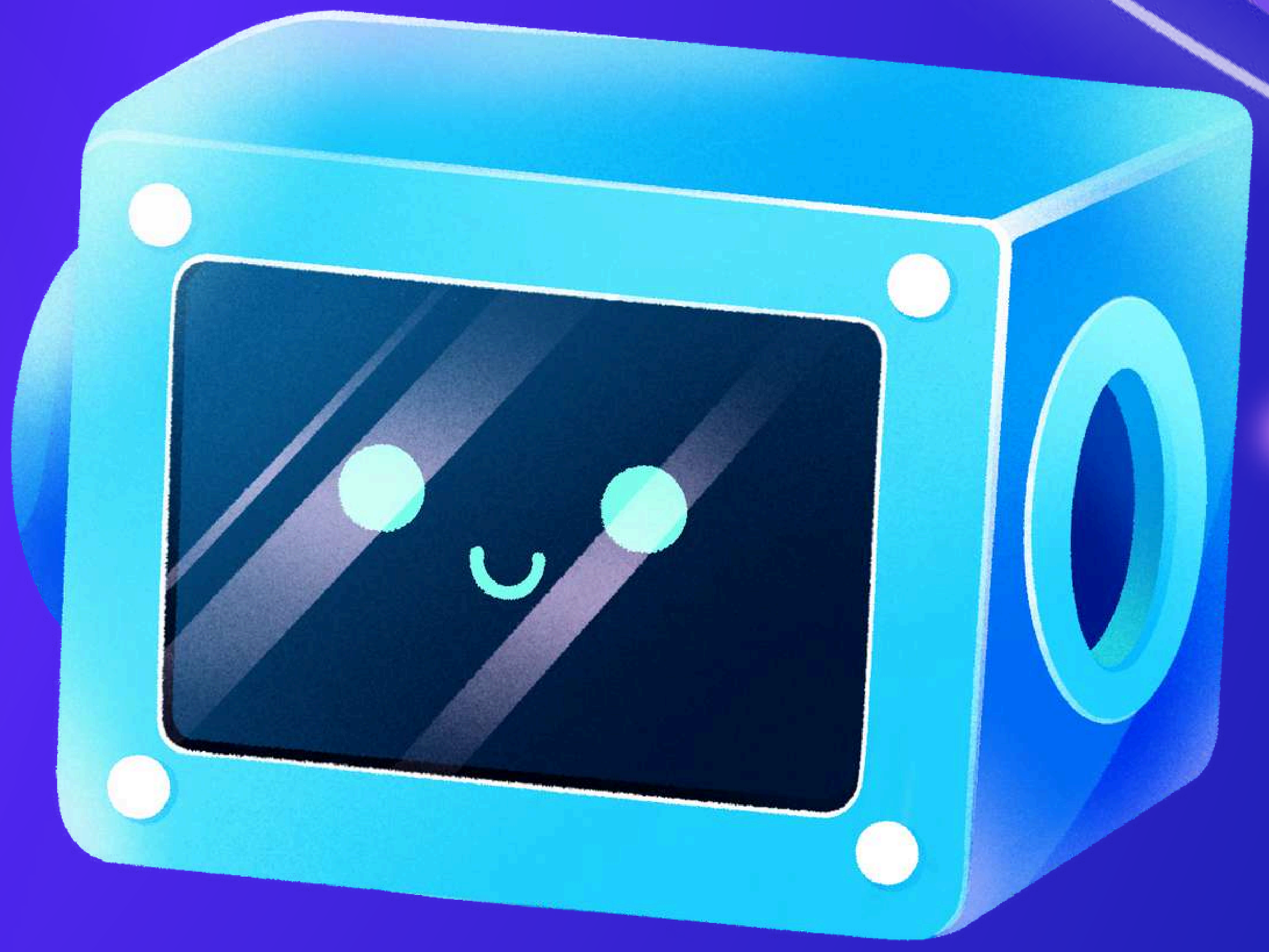
OBJECTIVE 02

Reduce recruiter effort by pre-filtering strong candidates.

OBJECTIVE 03

Visualize results to make decision-making faster and smarter.

PROJECT SCOPE



AUTOMATION PROCESS

01 Focuses on keyword matching using .txt format resumes.

02 Compares each resume to a single job description at a time.

03 Generates match scores, categorized results, and highlights top-matching resumes.

04 Currently excludes advanced ML-based semantic understanding.

RESULTS

01
- Each resume was compared with the job description using AI-powered techniques (TF-IDF and Cosine Similarity).

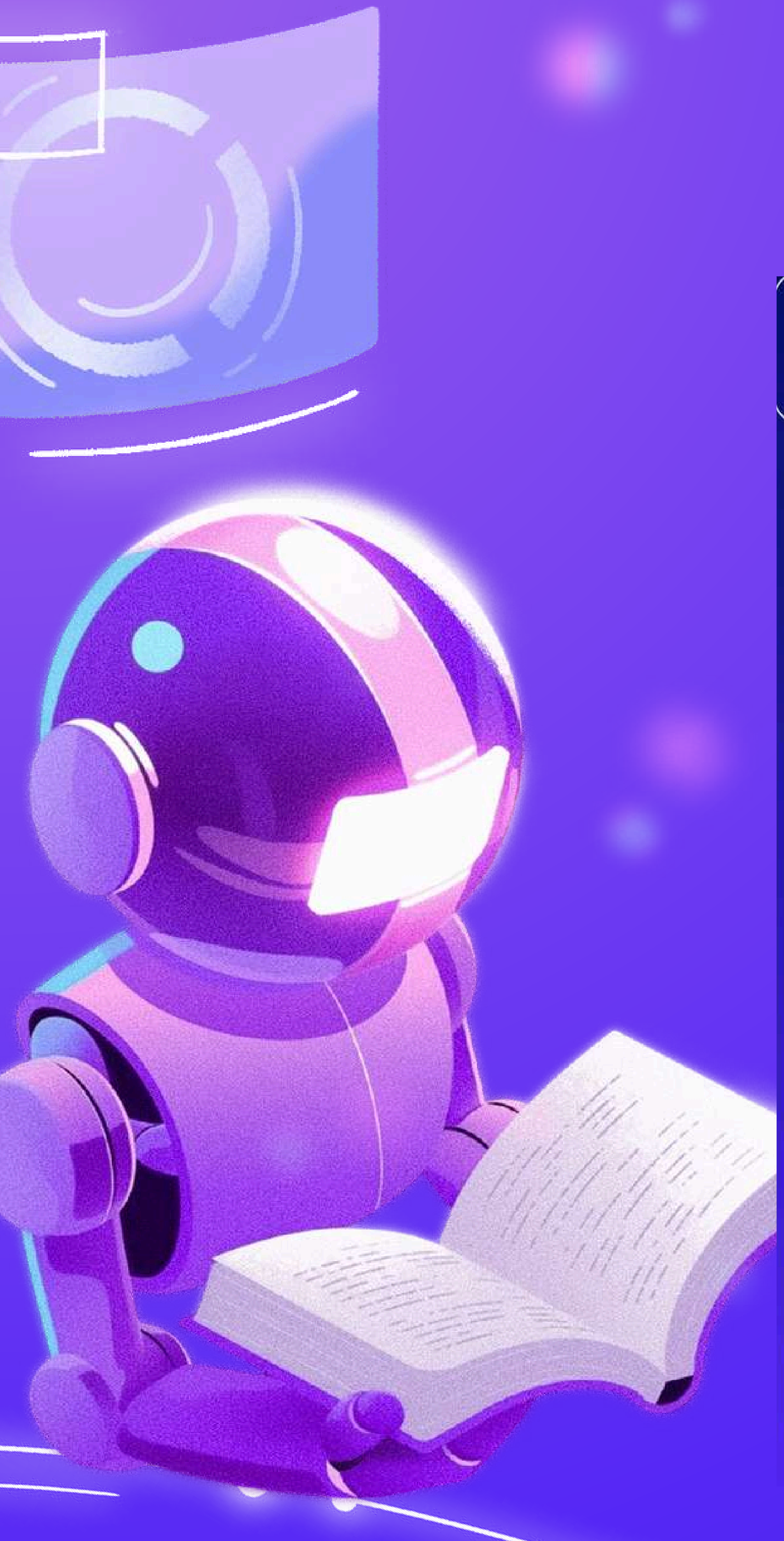
02
- The resume with the highest score matched 27.15% with the job description. A higher score indicates a stronger match with the job description.

```
results = []

for name, text in resume_texts.items():
    vectorizer = TfidfVectorizer()
    vectors = vectorizer.fit_transform([jd_text, text])
    score = cosine_similarity(vectors[0:1], vectors[1:2])[0][0]
    results.append((name, round(score * 100, 2)))

results
```

```
[('resume1.txt', 7.56),
 ('resume10.txt', 27.15),
 ('resume2.txt', 16.23),
 ('resume3.txt', 3.77),
 ('resume4.txt', 3.31),
 ('resume5.txt', 12.83),
 ('resume6.txt', 3.68),
 ('resume7.txt', 24.23),
 ('resume8.txt', 25.89),
 ('resume9.txt', 25.89)]
```



DASHBOARD

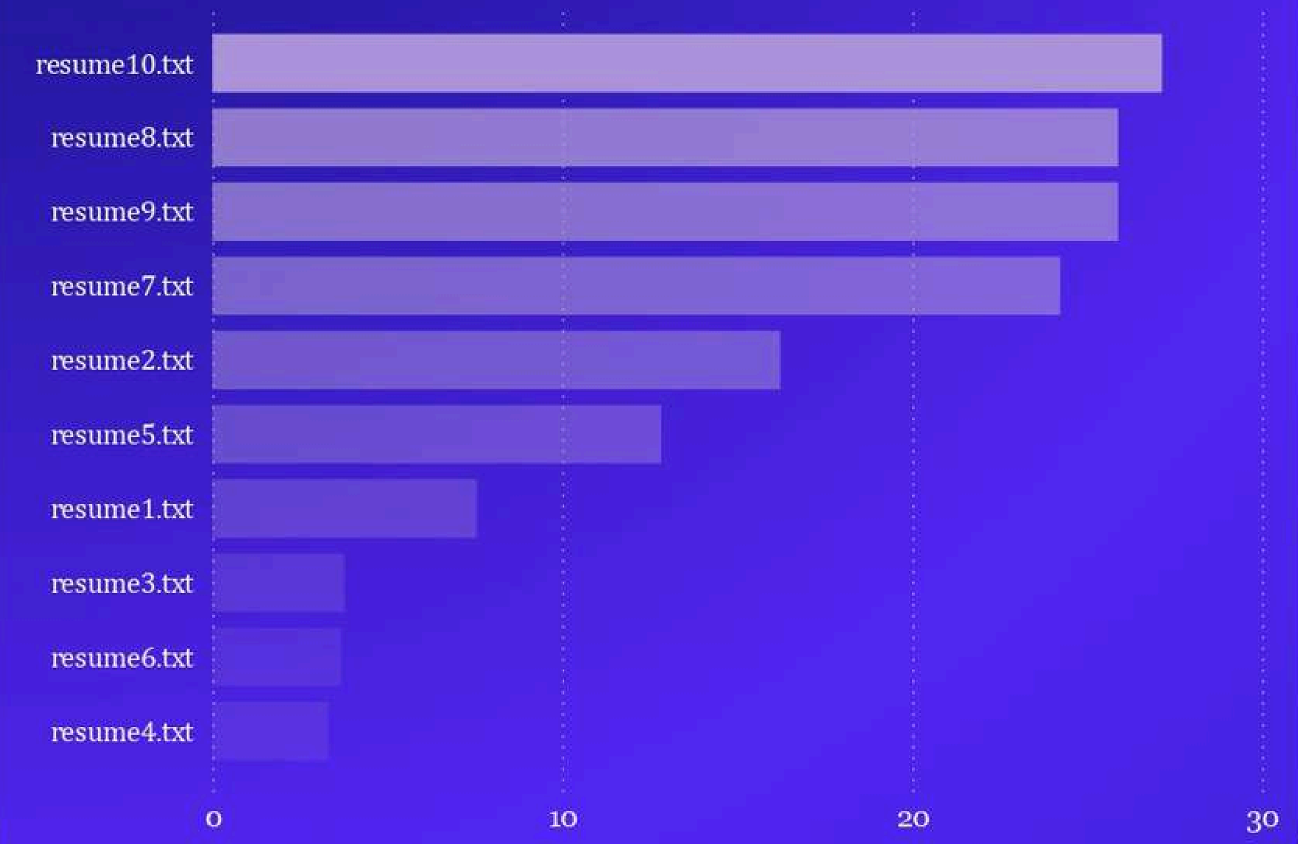
RESUME SCREENING BOT

10
TOTAL RESUMES

15.05
AVERAGE OF MATCH PERCENTAGE

3
HIGH MATCH COUNT

RESUME MATCHC BY CANDIDATE



resume10.txt
TOP RESUME

27.15
TOP MATCH SCORE

MATCH CATEGORY DISTRIBUTION

CONCLUSION

This AI-powered Resume Screening Bot uses Natural Language Processing (NLP) to automatically compare resumes with a Job Description. By applying TF-IDF and Cosine Similarity, it calculates how closely each resume matches the required skills and roles.

The results are visualized in Power BI, making it easy to:

- Identify top-matching candidates.
- Filter resumes based on match percentage
- Reduce time and effort in the shortlisting process.

This project showcases how AI can support recruiters making resume screening faster, smarter, and more data-driven — with no manual matching required.

THANK YOU!